WHAT IS GIVEN DECIDES

Which law to start with depends entirely on what information is available, and in particular on whether a complete side-angle pair (a, A) is given or immediately obtainable.

SOLVING A TRIANGLE

three of the six parts given

split by **whether a complete side-angle pair is available**

S

CASE 1

AAS, ASA, SSA

Start with the Law of Sines

a complete pair is given or immediately obtainable; watch for the ambiguous case with SSA

C

CASE 2

SAS, SSS

Start with the Law of Cosines

no complete pair exists until after the first computation

X

CASE 3

AAA, or fewer than three parts

Not solvable

AAA fixes the shape but not the size; fewer pieces cannot determine a triangle

IN COMBINATION

Many problems use both: a typical SAS problem finds the third side with the Law of Cosines, then switches to the Law of Sines for the remaining angles.